



Faculty of Technology & Engineering
Bachelor of Technology Programme
Computer Engineering
(B.Tech. CE)

ACADEMIC
REGULATIONS
&
SYLLABUS
(Choice Based Credit System)

Academic Year: 2024-2025

Chandubhai S. Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of Technology.

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives.

Department of Computer Engineering

Vision

To emerge as an eminent research driven entity in Education in the field of computer engineering to meet the needs of the fast paced modern society.

Mission

To develop competent and responsible Computer Professionals for amelioration of the society at large.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To prepare the student(s) for successful career as an engineer, a corporate or a government professional, a scientist, an academician, a technocrat, an administrator and an entrepreneur.
PEO2	To make students demonstrate their abilities to adapt to a rapidly changing environment by having learning approach and apply new skills and new technologies to solve the problems.
PEO3	To create an ambience where the students are cared for in every aspect and motivated to become excellent working professionals who will continue to cherish their association with the organization as a whole, staff and colleagues.
PEO4	To provide continued professional development and lifelong learning throughout their career to inculcate strong teamwork and by installing lacking skills for the benefit of the society.
PEO5	To prepare the students to apply their technical skill(s) to analyse and design appropriate solution(s) with social consciousness and ethical values.

PROGRAM ARTICULATION MATRIX

(-Not Matching,1-Low,2,3-High)

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
To provide world-class educational activities	3	3	3	3	3
To deploy latest technologies	-	2	3	3	-
To foster research activities in consultation with academia and industry	3	2	2	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO1	Apply good analytical, design and implementation skills required to formulate and solving computational problems.
PSO2	Excellent adaptability to function in multi-disciplinary work environment, good interpersonal skills in appreciation of professional ethics and societal responsibilities.

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B. Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be Minimum 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 University Examination

7.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.

7.2.2 **In order to earn the credit in a course a student has to obtain grade other than FF.**

7.3 Performance at Internal & University Examination

7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

7.3.2 A student failing to score 45% of the final examination will get a FF grade.

7.3.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Awards of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$\text{CGPA} \geq 7.5$
First class:	$\text{CGPA} \geq 6.0$
Second Class:	$\text{CGPA} \geq 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE):

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg.

Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non-Credit Course (NC) - AUDIT Course

A 'Non-Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non-Credit Courses

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 5	CE361	Operating System	3	2	5	4	30	70	25	25	150
	CE349	Theory of Computation	3	0	3	3	30	70	0	0	100
	CE362	Machine Learning	3	2	5	4	30	70	25	25	150
	CEXXX	Programme Elective-I	2	2	4	3	30	70	25	25	150
	CE363	Project-III	0	4	4	4	0	0	100	100	200
	CEXXX	Programme Elective-II	0	2	2	2	0	0	50	50	100
	CE346	Summer Internship-I	0	0	0	3	0	0	75	75	150
	HS131.02 A / HS142 A	Communication and Soft Skills / French	0	2	2	2	0	0	30	70	100
			11	14	25	25	120	280	330	370	1100
Sem 6	CE364	Cryptography & Network Security	3	2	5	4	30	70	25	25	150
	CE365	Compiler Construction	3	2	5	4	30	70	25	25	150
	CE366	Cloud Computing	3	2	5	4	30	70	25	25	150
	OCCEXXXX	Programme Elective-III	2	2	4	3	30	70	25	25	150
	CE367	Project-IV	0	4	4	4	0	0	100	100	200
	CEXXX	Programme Elective-IV	0	2	2	2	0	0	50	50	100
	HS132. 02 A	Contributory Personality Development	0	2	2	2	0	0	30	70	100
			11	16	27	23	120	280	280	320	1000

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives

- Programme Elective (PE):-
- Institute Elective (IE):-
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

List of Programme Electives			
Code	Programme Elective - I (PE - I)	Code	Programme Elective - II (PE - II)
CE384	Advanced Web Technology	OCCE3001	Deep Learning
CE385	Mobile Application Developments	OCCE3002	Big Data Computing
CE386	Advanced Java Programming	OCCE3003	Introduction to Internet of Things
CE387	.NET Core Programming	OCCE3004	Blockchain and its Applications
CE388	Embedded Systems	OCCE3005	Advanced Computer Network

Code	Programme Elective - I (PE - II)	Code	Programme Elective - II (PE - IV)
CE389	Introduction to Research Methodology	CE396	Research Writing and Ethics
CE390	Competitive Programming Essentials	CE397	Mastering Competitive Programming

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE361: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

Pre-requisite courses:

- Introduction to computer and computer organization.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Process Management	06
3.	Process Synchronization	06
4.	Deadlocks	04
5.	Memory Management	10
6.	Secondary Storage and Input Output Management	04
7.	File Systems	04
8.	System Protection	03
9.	Unix/Linux File System	04
10.	Epilogue	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction	02 Hours	05%
	What is an OS? OS structure, System Calls, Functions of OS, Evolution Of OS, OS design consideration for Multiprocessor and Multicore		
2.	Process Management	06 Hours	13%
	Process: What is Process? Process Control Block, Process States, Process API: fork(), exec() and wait() system calls, Process Scheduling, Threads: Types of Threads, Multithreading model, Scheduler activation		
3.	Process Synchronization	06 Hours	13%

	Race Conditions, Critical Section, Peterson's Solution, hardware primitives for synchronization, Mutex Locks, Semaphores, Monitors and Classical problems in synchronization(Producer-Consumer, Reader's & Writer Problem, Dining Philosopher Problem).		
4.	Deadlocks	04 Hours	9%
	Modelling, Characterization, Prevention and Avoidance, Detection and Recovery		
5.	Memory Management	10 Hours	22%
	Address Space, Address Translation, Multiprogramming with Fixed and variable partitions, Swapping, Segmentation, Free Space Management, Paging: Principle Of Operation, Page Allocation, H/W Support For Paging. Multilevel paging, Virtual Memory: Concept, Performance of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets.		
6.	Secondary Storage and Input Output Management	04 Hours	09%
	I/O Devices: Device Controllers, Device Drivers, Direct Memory Access. Goals of The I/O S/W, Interrupt Handler, Device Independent I/O Software Disks: Disk architecture, Disks Scheduling Algorithm, RAID		
7.	File Systems	04 Hours	09%
	File: File Naming, File attributes, File Operations, File Structure, File Types, Access Methods, Directories: Structure, Pathnames, Directory Operations. File System Implementation: partitioning and mounting, Allocation methods: Contiguous Allocation, Linked Allocation, Indexed Allocation.		
8.	System Protection	03 Hours	07%
	Goals of protection, Principles of protection, Domain of protection, Access matrix, Implementation of Access matrix, Access control		
9.	Unix/Linux File System	04 Hours	09%
	Buffer Cache, Inodes, The system calls - ialloc, ifree, namei, alloc and free. Mounting and Unmounting , files systems, Network File systems. EXT file system in linux		
10.	Epilogue	02 hours	04%
	Pointers to advanced topics: distributed OS, multimedia OS, embedded OS, real-time OS, OS for multiprocessor machines, research problems in OS		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Analyze the basic structure of OS and architectural components involved in OS design
CO2	Evaluate the performance of various process scheduling algorithms after understanding various process states and its mechanism to manage
CO3	Evaluate the requirements of process synchronization and coordination in contemporary operating system.
CO4	Compare the various mechanisms to manage memory by applying memory management algorithms and schemes along with management of input/output devices
CO5	Compare the various mechanisms to manage files by applying file allocation methods along with security aspects
CO6	Analyze the administration of Linux operating system

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	2	-	-	-	-	-	-	1	-	-
CO2	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO3	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO4	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO5	2	3	-	-	2	-	-	-	-	-	-	-	1	-
CO6	2	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley
2. Operating Systems Internals and Design Principles, William Stallings, Seventh Edition, Prentice Hall

❖ **Reference book:**

1. Operating Systems: Three Easy Pieces by Remzi Arpaci-Dusseau and Andrea Arpaci-Dusseau (available online for free)
2. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
3. Operating Systems, D.M. Dhamdhare, TMH
4. Unix System Concepts & Applications, Sumitabha Das, TMH
5. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

❖ **Web material:**

1. NPTEL course: <https://nptel.ac.in/courses/106/106/106106144/>
2. Simulator: <https://cpudemo.azurewebsites.net/#/>

❖ **Software:**

1. Simulator: OS Sim (OS Concepts Simulator)

CE349: THEORY OF COMPUTATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	-	-	3	3
Marks	100	-	-	100	

Pre-requisite courses:

- Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithm (Computational Complexity)

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction	03
2	Mathematical Terms and Theory	05
3	Regular Grammar & Languages, Regular Expression, Finite Automata	14
4	Context Free Grammar & Languages, Push down	13
5	Turing Machine, Recursively Enumerable Languages	08
6	Decidable & Undecidable Problems	02
	Total hours (Theory) :	45
	Total hours (Lab) :	00
	Total hours :	45

Detailed Syllabus:

1.	Introduction	03 Hours	06 %
	Alphabet, String, Language, Formal Grammar, Chomsky Hierarchy, Introduction to Automata		
2.	Mathematical Terms and Theory	05 Hours	10 %
	Mathematical Inductions, Recursive Definitions		
3.	Regular Grammar & Languages, Regular Expression, Finite Automata	14 Hours	32 %

	Regular Language, Regular Expressions, Applications, Chomsky Hierarchy, Finite Automata, Nondeterministic Finite Automata, Kleen's Theorem, Automata with Output (Moore Machine, Mealy Machine), Properties of Regular Languages (Pumping Lemma, Closure Property, Decision Algorithm)		
4.	Context Free Grammar & Languages, Push down Automata	13 Hours	30%
	The Chomsky, Notion of Grammars and Languages Generated by Grammars, CFG, CFL, Regular Language and Regular Grammar, Derivation Tree and Ambiguity, BNF, CNF, GNF, CFL properties (Pumping Lemma, Closure Property, Decision Algorithm), Intersections and Complements of CFL, Non-CFL, Definition, DPDA, NPDA, Equivalence of CFG and PDA		
5.	Turing Machine, Recursively Enumerable Languages	08 Hours	18%
	Definition, Model of Computation, Combining TM, Variations of TM, Non Deterministic TM, Universal TM, Recursively Enumerable and Recursive, Enumerable Languages, Context sensitive languages		
6.	Decidable & Undecidable Problems	02 Hours	04 %
	Tractable and Intractable Problems, Complexity Classes, Tractable and Possibly Intractable Problems, P and NP Completeness, Countability		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply basic concepts of theory of computation in the computer field in order to solve computational problems.
CO2	Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.

CO3	Analyze and design finite automata, pushdown automata and Turing machine for formal languages.
CO4	Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO5	Identify limitations of some computational models and possible solutions.
CO6	Design context free grammars for formal languages.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	2	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-	3	-
CO4	2	2	2	-	-	-	-	-	-	-	-	-	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Introduction to Languages and Theory of Computation, John C. Martin, TMH

❖ Reference book:

1. An introduction to automata theory and formal languages, Adesh K. Pandey, S. K. Kataria & Sons
2. Introduction to computer theory, Deniel I. Cohen, John Wiley & Sons Inc
3. Computation: Finite and Infinite, Marvin L. Minsky, Prentice-Hall
4. “An introduction to Formal Languages and Automata”, Peter Linz, 6th edition, Jones & Bartlett Learning

5. “Introduction to the Theory of Computation”, Michael Sipser, 3rd edition, Cengage Learning.

❖ **Web material:**

1. https://www.youtube.com/playlist?list=PLEbnTDJUr_IdM___FmDFBz0zCsOFxfK
2. <http://nptel.ac.in/courses/106103070/>
3. <http://nptel.ac.in/courses/106104028/>
4. <http://nptel.ac.in/courses/106106049/>
5. <https://www.youtube.com/watch?v=4GLC-s0PQLY>

CE362: MACHINE LEARNING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Probability, linear algebra, calculus and programming language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Data cleaning and Preprocessing	06
2.	Supervised Machine Learning	14
3.	Unsupervised Machine Learning	11
4.	Ensemble learning	04
5.	Deep Neural Networks	10
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Data Cleaning and Pre-processing	06 Hours	13%
	Handling Missing data, Data Exploration and Visualization, Handling Outliers, Data Integration and Aggregation, Feature selection: Filter Methods, Wrapper Methods, Embedded Methods, Data Transformation: Feature scaling and normalization, Encoding categorical variables: one-hot encoding, label encoding, Data transformation for skewed distributions		
2.	Supervised Machine Learning	14 Hours	31%
	What is supervised ML, Linear Regression for univariate and Multivariate data, Cost function, Gradient Descent, Logistic regression, Under fitting and Over fitting, Support Vector		

	Machine, Decision Tree, Random Forest, Artificial Neural Network architecture, Activation functions, Forward pass in ANN, Back propagation in ANN, Model Evaluation techniques.		
3.	Unsupervised Machine Learning	11 Hours	25%
	What is unsupervised ML, Distance Techniques: Euclidean Distance, Manhattan Distance, Minkowski Distance, Cosine Similarity, Hamming Distance. Different clustering algorithm: K-means, Hierarchical Clustering, DBSCAN, Evaluate clustering algorithm, Evaluation Techniques: Evaluate Clustering metrics - Silhouette Score, Calinski-Harabasz Index, Davies-Bouldin Index, Dunn Index. Dimensionality reduction: Principal Component Analysis(PCA)		
5.	Ensemble learning	04 Hours	09%
	Gradient Boosting, XGBoost (Extreme Gradient Boosting), AdaBoost (Adaptive Boosting)		
5.	Deep Neural Networks	10 Hours	22%
	Introduction to Convolutional Neural Networks, Convolutional Neural Network Layers, CNN Architectures and Variants, Introduction to Recurrent Neural: Understanding sequential data and its challenges, Anatomy of a Recurrent Neural Network (RNN), vanishing gradient problem Networks, Long Short-Term Memory (LSTM) Networks.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Addressing data quality issues, effectively cleaning and preprocessing raw datasets, and preparing them for machine learning algorithm.
CO2	Capable of building predictive models for classification and regression tasks, and skilled in selecting, training, and evaluating machine learning algorithms on labeled data

CO3	Solid understanding of unsupervised learning methods, including clustering and dimensionality reduction
CO4	Proficient in implementing and optimizing gradient boosting models for both regression and classification tasks
CO5	Understand the principles and architecture of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) Networks.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	1	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO3	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO4	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO5	3	-	3	2	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 2nd Edition, Aurelien Géron, O'Reilly
2. Machine Learning Yearning, Andrew Ng, GitHub; eBook (Draft, 2018); eBook (MIT Licensed)
3. Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville, The MIT Press (17 December 2016)

❖ Reference book:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification. Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.

❖ **Web material:**

1. <https://www.youtube.com/watch?v=foHSmB48rY&list=PLKvX2d3IUq586Ic9gIhZj6ubpWV-OJfl4>
2. https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6EcXTbcX1uRg2_u4xOEky0
3. <https://www.youtube.com/watch?v=UzxYlbK2c7E>
4. <https://www.youtube.com/playlist?list=PLAwxTw4SYaPkQXg8TkVdIvYv4HfLG7SiH>
5. [Recommended online course:](#)
6. [Coursera:https://www.coursera.org/specializations/machine-learning-introduction](https://www.coursera.org/specializations/machine-learning-introduction)
7. [Coursera: https://www.coursera.org/specializations/deep-learning](https://www.coursera.org/specializations/deep-learning)

❖ **Software:**

1. Anaconda
2. GoogleColab

CE384: ADVANCED WEB TECHNOLOGY(PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	0	4	3
Marks	100	50	0	150	

Pre-requisite courses:

- HTML, CSS, JavaScript

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	JavaScript Essentials	03
2.	JavaScript Modules and Object Oriented programming	02
3.	Asynchronous JavaScript	03
4.	Learning about the Internals of Node.js	03
5.	Working with ExpressJs and Form Handling	06
6.	Authentication, Authorization and Security	03
7.	MongoDB	05
8.	Sequelize ORM Framework	05
	Total hours (Theory) :	30
	Total hours (Lab) :	30
	Total hours :	60

Detailed Syllabus:

1.	JavaScript Essentials	03 Hours	10%
	Variables, Primitive data types, Analyzing and modifying data types, Operators, JavaScript Multiple Values, Arrays and their properties, Array methods, Multidimensional arrays, Objects in JavaScript, Working with objects and arrays, Basic functions, Parameters and arguments, Special functions and operators, Returning function values, Variable scope in functions, Recursive functions, Nested functions, Anonymous functions		
2.	JavaScript Modules and Object Oriented programming	02 Hours	07%
	Object-oriented programming, Classes and objects, Classes		

	Inheritance, Prototypes, Modules with Classes, Objects, and Closures, Modules in ES6: ES6 Exports, ES6 Imports, Imports and Exports with Renaming, Re-Exports, JavaScript Modules on the Web, Dynamic Imports with import(), Iterators and Generators: yield* and Recursive Generators, Advanced Generator Features		
3.	Asynchronous JavaScript	03 Hours	10%
	Asynchronous Programming with Callbacks, Promises, async and await, Asynchronous Iteration		
4.	Learning about the Internals of Node.js	03 Hours	10%
	NodeJs Architecture, understanding event Loop, Package manager: npm and yarn, Create and publish custom package Understanding Requests, Sending Responses, Routing Requests, Redirecting Requests, Parsing Request Bodies Understanding Event Driven Code Execution, Blocking and Non-Blocking Code		
5	Working with ExpressJs and Form Handling	06 Hours	19%
	Adding Middleware, How Middleware Works? Express.js - Looking Behind the Scenes, Handling Different Routes, Parsing Incoming Requests, Limiting Middleware Execution to POST Requests, Using Express Router, Adding a 404 Error Page, Filtering Paths, Model View Controller Working with Forms, Validation and sanitization, Form Handling with ExpressJS, Using Fetch to Send Form Data, Sending Email and File Upload and Download		
6	Authentication, Authorization and Security	03 Hours	10%
	Externalizing Credentials, Cookies in Express, Examining Cookies, Sessions: Memory Stores, Using Sessions, Using Sessions to Implement Flash Messages, JSON Web Token(JWT), Rate Limiting		
7	MongoDB	05 Hours	17%
	Overview of MongoDB Fundamentals - the document model, how to create, update, read, and delete documents, indexing, and also learn about MongoDB Atlas, MongoDB databases, collections, and documents, Connecting to a MongoDB Database, MongoDB CRUD Operations: Replace and Delete Documents, Modifying Query		

	Results, MongoDB Aggregation, MongoDB Indexes, Data Modeling, Transaction.		
8	Sequelize ORM Framework	05 Hours	17%
	Defining and Using Sequelize Models: Defining models for a database, Exploring the various Sequelize data types and when to use them, Migrating schematic changes and data from Sequelize to the database, Manipulating and querying data using Sequelize, Validating, Customizing, and Associating Your Data, Validating Models, Associating Models, Implementing Transactions with Sequelize		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	How to use Express JS framework with Node.js with real projects with request (Understand)
CO2	Handle Form, session and cookies.
CO3	Render Client-side/Server-side website using template engine (Apply)
CO4	Create CRUD operations with MongoDB and Mongoose (Create)
CO5	Build a fast, scalable, feature-rich RESTful API (includes filters, sorts, pagination, and much more) (Create)
CO6	Build a complete, beautiful & real-world application from start to finish (API and server-side rendered website)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	3	2	3	1	-	2	1	1	3	3	1	1
CO2	-	1	3	2	3	1	-	1	1	1	3	3	1	1
CO3	-	1	3	2	3	1	-	1	2	2	3	3	2	1
CO4	-	1	3	2	3	2	-	1	2	2	3	3	3	1
CO5	-	1	3	2	3	2	-	1	2	3	3	3	3	1
CO6	-	1	3	2	3	3	-	2	3	3	3	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Web Development with Node and Express, 2nd Edition by Ethan Brown, O'Reilly Media

❖ **Reference book:**

1. Node.js Web Development - Fifth Edition By David Herron, Packtpub
2. Node Cookbook - Fourth Edition By Bethany Griggs, Packtpub

❖ **Web material:**

1. <https://nodejs.org/en/docs/>
2. <https://handlebarsjs.com/guide/>

❖ **Software:**

1. Node.js, Express.js, MondoDB
2. IDE: Visual code Editor, Atom, Sublime

CE385: MOBILE APPLICATION DEVELOPMENTS (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- Basic Design concept with XML, Database management system.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Kotlin	02
2.	Functional Programming	03
3.	Objected oriented programming	05
4.	Dart Programming	06
5.	Flutter User Interface (Flutter UI)	06
6.	Theming, Styling and Routing	08
	Total hours (Theory) :	30
	Total hours (Lab) :	30
	Total hours :	60

Detailed Syllabus:

1.	Introduction to Kotlin	02 Hours	6%
	Introduction, Why Kotlin? Functions, Looping & Ranges, Objects Everywhere, Creating Class, Constructors, Packages, Exceptions, Enumerations, Data Classes, Safe Calls & the Elvis Operator, Introduction to Generics.		
2.	Functional Programming	03 Hours	10%
	Lambdas, Higher-Order Functions, Lists, Maps, Recursion		
3.	Objected oriented programming	05 Hours	17%
	Interfaces, Secondary Constructors, Inheritance, Abstract Classes, Upcasting, Polymorphism, Composition, Inheritance & Extensions,		

	Class Delegation, Down casting, Nested Classes, Objects, Inner Classes, Companion Objects		
4.	Dart Programming	06 Hours	20%
	Dart Introduction, Stagehand –project generator, The pubspec file, Introducing async programming with features		
5.	Flutter User Interface (Flutter UI)	06 Hours	20%
	Introduction to Flutter, widgets introduction, hello flutter, Understanding built-in widgets, Creating UI with widgets, creating custom widgets		
6.	Theming, Styling and Routing	08 Hours	27%
	Themes widgets, Using custom fonts, Dynamic styling with MediaQuery, Understanding the navigator widget. Names routes, Screen transitions, Hero animations.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Knowledge of Kotlin programming language: Solid understanding of the Kotlin programming language.
CO2	Understanding of flutter framework: Through understanding of the flutter framework and its architecture.
CO3	Proficiency in developing mobile application: Developing mobile applications using the flutter and Kotlin
CO4	Ethical consideration: Understand the ethical considerations involved in mobile application development

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-
CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Bruce Eckel and Svetlana Isakova, Atomic Kotlin, Mindview LLC.
2. Alessandro Biessek, Flutter for Beginners, Packt Publishing Limited

❖ **Reference book:**

1. "Android Programming: The Big Nerd Ranch Guide" by Bill Phillips, Chris Stewart, and Kristin Marsicano
2. "iOS Programming: The Big Nerd Ranch Guide" by Christian Keur and Aaron Hillegass
3. "Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart" by Alessandro Biessek

❖ **Web material:**

1. <https://kotlinlang.org/>
2. <https://flutter.dev/>

❖ **Software:**

1. Android Studio
2. Xcode
3. Visual Studio

CE386: ADVANCED JAVA PROGRAMMING (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- Object oriented Language - JAVA
- DBMS
- Web Programming

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Hibernate Introduction	02
2.	The Persistence Life Cycle	03
3.	Mapping with Annotations	05
4.	Hibernate Query Annotations	05
5.	Getting Started with Spring Boot	03
6.	Building REST APIs Using Boot	04
7.	Building Microservice using Spring Boot	08
	Total hours (Theory) :	30
	Total hours (Lab) :	30
	Total hours :	60

Detailed Syllabus:

1.	Hibernate Introduction	02 Hours	6%
	Origins of Hibernate and Object/Relational Mapping, Hibernate as a Persistence Solution, A Hibernate Hello World Example, The Steps Needed to Integrate and Configure Hibernate, Understanding Where Hibernate Fits into Your Java Application, Writing, reading and Updating		

	data,		
2.	The Persistence Life Cycle	03 Hours	10%
	Introducing the life cycle, Entities, Classes and Names, Identifiers, Entities and Associations, Saving Entities, Object Equality and Identity, 'Loading, Refreshing, Merging, Updating and Deleting Entities', Cascading Operations, Lazy Loading, Proxies, Collection Wrappers.		
3.	Mapping with Annotations	05 Hours	17%
	Primary keys, Lazy Loading, Associations (One-to-one, One-to-Many and Many-to-One, Many-to-Many), Specification of database (Column types and Sizes), Cons and Pros of annotation, JPA 2 Persistence Annotations (Entity Beans, Primary keys, Computing Primary Keys, inheritance etc.), Configuring the annotated classes, Hibernate-specific Persistence Annotations (immutable, Natural IDs).		
4.	Hibernate Query Language	05 Hours	17%
	Syntax Basics (Update, Delete, Insert, Select), Logging and Commenting the Underlying SQL, Using restrictions with HQL, Using named parameters, Using Criteria API (Using restrictions with criteria, paging through the result set, obtaining a unique result, sorting the query's result), When to use filters, Defining and attaching Filters (Filters with annotations, Filter with XML Mapping Document).		
5.	Getting Started with Spring Boot	03 Hours	10%
	Working with JDBC, Working with JPA (Overview of Spring Data JPA, JPA Repository, Configure the Entity, Create the Repository, Modify and Autowire the repository into the service).		
6.	Building REST APIs Using Spring Boot	04 Hours	13%
	What is REST API, Connecting spring boot to the project, Connecting spring boot to the database, Creating user model, Creating Controller		
7.	Building Microservice using Spring Boot	08 Hours	27%

	What are microservices, Why build Microservices, Microservices Architecture, Microservice resilience with Spring Cloud, Build streaming data microservices with Spring Cloud Stream,		
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Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Illustrate database access and details for managing information using the JDBC API
CO2	Design to map java classes and object associations to relational database tables with hibernate framework.
CO3	Demonstrate web application lifecycle using servlet, and JSP.
CO4	Identify the user interface control and build custom tag library using JSF
CO5	Develop real-world applications using java frameworks and understands about MVC architecture with its application.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-
CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-
CO5	2	3	3	2	3	-	-	-	-	-	-	-	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Joseph B. Ottinger, Jeff Linwood and Dave Minter Begging Hibernate, Apress Publications.
2. Beginning Spring Boot 3, K. Siva Prasad Reddy, Sai Upadhyayula, Apress Publications.
3. Spring Microservices in Action, John Carnell, Manning Books.

❖ **Reference book:**

1. JDBC™ API Tutorial and Reference, Third Edition, Maydene Fisher, Jon Ellis, Jonathan Bruce, Addison Wesley
2. Beginning JSP, JSF and Tomcat, Giulio Zambon, Apress
3. JSF2.0 CookBook, Anghel Leonard, PACKT publication

❖ **Web material:**

1. http://www.service-architecture.com/application-servers/articles/j2ee_web_site_architecture.html
2. <http://www.oracle.com/technetwork/java/javaee/overview/index.html>
3. <http://www.roseindia.net/struts/hibernate-spring/index.shtml>

❖ **Software:**

1. JDK 8+
2. NetBeans IDE
3. Eclipse
4. IntelliJ IDEA

CE387: .NET CORE PROGRAMMING (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- Object Oriented Concepts
- Basic programming knowledge

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to .NET framework and ASP.NET	03
2.	Developing applications with ASP.NET	06
3.	Database connectivity in ASP.NET	07
4.	MVC architecture in .NET framework	04
5.	Introduction to .NET Core framework	05
6.	Advanced concepts in ASP.NET	05
	Total hours (Theory) :	30
	Total hours (Lab) :	30
	Total hours :	60

Detailed Syllabus:

1.	Introduction to .NET framework and ASP.NET	03 Hours	10%
	Introduction to .NET framework, Introduction to Web Programming, Introduction to ASP and ASP.NET, Deploying ASP.NET Application. Familiarity with the .NET Framework architecture, including the Common Language Runtime (CLR), Base Class Library (BCL), and Common Type System (CTS). Knowledge of .NET languages such as C# .NET and their syntax, features, and capabilities. C# Language Context, Classes & Objects, OOPS basics , Arrays, Collections & Generics, Encapsulation, Inheritance, Polymorphism, Data Abstraction, Namespace, Nested Namespace ,Delegates & Events , Properties, Indexer & Indexer Overload, Errors and Exceptions, Strings, Regular Expressions ,Collections, Reflection, Class Libraries (DLL).		

2.	Developing applications with ASP.NET	06 Hours	20%
	ASP.NET Page Life Cycle, Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Developing a web form, Working with master pages, State management techniques, Application Tracing, Error Handling		
3.	Database connectivity in ASP.NET	07 Hours	24%
	Introduction to ADO and ADO.NET, ADO.NET architecture, Connection Oriented and Connection Less architecture, Binding data to web controls and working with data controls, working with XM. Knowledge of ADO.NET objects such as SqlConnection, SqlCommand, SqlDataAdapter, and DataSet.		
4.	MVC architecture in .NET framework	04 Hours	12%
	Introduction to MVC Architecture in ASP.NET, CRUD operation using MVC architecture, working with web controls in MVC architecture, introduction to LINQ. Understanding of routing, controllers, models, and views in ASP.NET MVC applications. Ability to create and configure controllers, define routes, and implement views using Razor syntax.		
5.	Introduction to .NET Core framework	05 Hours	17%
	Introduction to ASP.NET Core, Introduction to .NET core, characteristics of .NET core, difference between .NET and .NET core ASP.NET Core Features, Advantages of ASP.NET Core, MVC Pattern, Understanding ASP.NET Core MVC Model, Views and Controllers in ASP.NET Core, ASP.NET Core Environment Setup, ASP .NET Core First Application, Project Layout Controllers & Action Methods, Helpers, CRUD operation with .Net Core. Introduction to service oriented architecture concept, Web API Development and Consume Web API, SOAP protocol, Concept of UDDI and WSD or REST.		
6.	Advanced concepts in ASP.NET Core: Entity Framework	05 Hours	17%
	Introduction to ASP.NET AJAX, AJAX control toolkit and extender controls. Entity Framework, Entity Framework Features, Database-First, Code-First, Model-First Approach, Entity Framework core, CRUD using Entity Framework , Introduction to NuGet.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Students will be able to understand .NET framework, ASP.NET technologies and develop simple applications using the framework. C# Object oriented concepts understanding.
CO2	Students will be able to develop database centric application using ADO .Net.
CO3	Students will be familiar with concepts of MVC for application development.

CO4	Students will able to understand SOA concepts and develop simple SOA based applications with .NET framework. Students will able to understand .Net Core Framework and develop cross platform applications with .NET Core framework.
CO5	Students will be familiar with the concepts of AJAX and other advance .net technology concepts. Students will be familiar with the concepts of Entity Framework and Entity Frame Work Core along with NuGet.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	3	-	2	1	-	-	2	-	-	2	1
CO2	2	3	2	2	-	2	1	-	-	2	2	1	3	1
CO3	1	2	2	-	-	-	-	-	-	-	2	1	3	2
CO4	2	3	3	-	2	-	-	-	-	-	2	1	3	2
CO5	3	2	3	2	-	-	3	-	-	-	-	2	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

Core Books:

1. Cogent Solutions Inc.: ASP.Net 4.5 Black book, Dreamtech press, 2019
2. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2017
3. Stephon Walther: ASP.Net Unleashed, BPB publication, 2018

Reference Books:

1. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming, Developer's Guide, Dreamtech, 2016.
2. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2016.
3. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2019 internals and troubleshooting, Wiley India publication, October, 2019.

Web Materials/ Reading Material:

- <http://msdn.microsoft.com/en-us/aa336522.aspx> [For software download]
- <https://docs.microsoft.com/en-us/aspnet/overview> [For ASP.NET technical documentation]
- https://www.tutorialspoint.com/dotnet_core/dotnet_core_overview.htm [for .NET Core]

CE388 : EMBEDDED SYSTEM (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics
- Microprocessor and Computer Architecture
- Basic programming knowledge

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of Embedded System:	04
2.	8-bit Microcontrollers Architecture (Atmega 8, AVR)	10
3.	Interface Digital and Analog I/O Devices (Arduino Interfacing):	04
4.	Real-Time Operating Systems	08
5.	Embedded system Applications (Arduino)	04
	Total hours (Theory) :	30
	Total hours (Lab) :	30
	Total hours :	60

Detailed Syllabus:

1.	Overview of Embedded System	04 Hours	14%
	Embedded system: Embedded System Characteristics Examples: Washing Machine, Chocolate Vending Machine, Room Temperature Controller, Operating System (OS): Types of OS, Types of Mobile OS, Characteristics of Real Time Operating System		
2.	8-bit Microcontrollers Architecture (Atmega 8, AVR)	10 Hours	30%
	Microcontroller Types: PIC, AVR, ARM: features and applications AVR microcontroller: Types, Internal Architectural, Block diagram of controller (Atmega 8), Functions of each pins of Atmega 8, Configuration of Timers and Counters. Interrupts, ADC Working, Essential Peripheral circuits: Crystal Circuit, Power supply, Oscillator Circuit, Initial programming configurations of Atmega8: port, counter, timer, Bootloader Circuit, ISP of Atmega 8,		
3.	Interface Digital and Analog I/O Devices (Arduino Interfacing):	04 Hours	14%

	Basic Interfacing and I/O Concept Interfacing LED, Switch, 7seg LED its and Code Interfacing POT, LM35, Accelerometer (ADXL3C5C) and its Code, Interfacing keypad, Initialization for serial port, Interfacing DC motor, Interfacing 16x2 LCD.		
4.	Real-Time Operating Systems	08 Hours	28%
	RTOS in embedded systems, RTOS scheduling models, task enrolment and scheduling, task prioritization, context switching, multitasking, pre-emptive and cooperative inter-task communication, event management; locking mechanism, interrupt handling, Introduction to Open Source RTOS like uCOS, Free RTOS, etc. and overview of Mobile OS		
5.	Embedded system Applications (Arduino):	04 Hours	14%
	Motor Driver L293D, IR Sensor, Interfacing L293D with Arduino, Line Follower Robot, Home automation. Case Study based on Recent Trends in Architecture and Applications		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the fundamental skills knowledge of embedded system, different examples, its characteristics
CO2	Understand the different processors architecture and able to do a programming for Atmega 32 Processor.
CO3	Analyse the requirement of interfacing peripheral devices with Microcontroller
CO4	Understand operating system and do basic programming of real time operating system
CO5	Analyse the systems requirements to meet the specifications

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	1	-	-	-	2	-	-
CO2	3	2	-	-	-	-	-	-	2	1	-	1	1	-
CO3	3	3	2	-	1	-	-	-	2	1	-	1	1	-
CO4	3	1	1	-	1	-	-	-	2	1	-	1	1	-
CO5	3	3	3	2	3	3	2	2	3	1	1	2	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**Reference Books:**

1. Embedded System: Architecture, Programming and Design by Rajkamal, TMH
2. ARM System on Chip Architecture by Steve Furber, Pearson Education
3. Computer as Components: Principles of Embedded Computing System Design, Wayne Wolf, Morgan Kaufmann Publication
4. AVR Microcontroller and Embedded Systems: Using Assembly and C, Mohammad Ali Mazidi, Sarmad Naimi, Sepehr Naimi, Pearson Publication

Web Materials/ Reading Material:

1. An Embedded Software Primer Addition, by David E. Simon ,Wesely.
2. System Design: A unified Hardware/Software introduction, by Frank Vahid & Tony D.Givargis, Addison Wesley 2002
3. Embedded Systems Architecture, by Tammy Noergaard Elsevier,
4. Fundamentals of Embedded Software: Where C and Assembly Meet, by Daniel Lewis, Prentice Hall
5. Embedded C programming and the ATMEL AVR by Barnett, cox and o’cull, Thomson

CE363: PROJECT-III

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	4
Marks	-	200	-	200	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - Project Synopsis
 - Software Requirement Specification
 - SPMP
 - Final Project Report
 - Project Setup file with Source code
 - Project Presentation (PPT)

- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 60

Total hours : 60

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	An ability to function effectively in teams to accomplish a common goal.
CO2	An ability to apply knowledge of computing and engineering to evaluate project requirements.
CO3	An ability to design, implement and evaluate a computer-based system, process, component, or program to meet desired needs.
CO4	An ability to analyze the local and global impact of computing on individuals, organizations, and society.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	An ability to communicate effectively with end user.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	2	-	1	2	-
CO2	3	-	1	2	-	-	-	-	-	-	-	-	3	-
CO3	-	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	-	-	-	-	3	2	-	-	1	-	-	2	-
CO5	-	-	-	-	-	-	1	-	-	2	-	3	-	2
CO6	-	-	-	-	-	-	3	1	-	-	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines ,Journals & online course platforms of related topics

❖ Web material:

1. www.sciencedirect.com
2. www.elsevier.com
3. <https://www.udemy.com/>
4. <https://www.udacity.com/>
5. <https://nptel.ac.in/course.html>
6. <https://www.futurelearn.com/>

❖ Software:

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS

CE389: INTRODUCTION TO RESEARCH METHODOLOGY (PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Pre-requisite courses:

Basic Mathematics and Statistics, Subject-Specific Knowledge, Computer Literacy.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Research	06
2.	Literature Review	10
3.	Technical Writing	08
4.	Tools / Techniques for Research	06
	Total hours (Theory) :	-
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Research	06 Hours	20%
	Meaning, Objective and Motivation in Research. Types of Research: Descriptive vs. Analytical, Applied vs. Fundamental, Quantitative vs. Qualitative, Conceptual vs. Empirical, Concept of Applied and Basic Research Process, Criteria of Good Research.		
2.	Literature Review	10 Hours	34%
	Objectives of Review of Literature, Importance of Literature Review in Defining a Problem, Steps in Conducting Literature Review - Guidelines on Style, Mechanics and language usage - Writing up to the literature Reviewed - Some Examples		

3	Technical Writing	08 Hours	26%
	Introduction, Steps of Research, Types of Research Article, Structure of Review Research Article, Review Article, Publication Ethics, Length of Review Article		
4.	Tools for Research	06 Hours	20%
	Methods to search required information effectively, Reference Management Software like Zotero /Mendeley, Software for paper formatting like LaTeX / MS Office, Overleaf.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Explain the fundamental concepts and principles of research methodology, including the scientific method and various research paradigms.
CO2	Design a comprehensive research plan, including selecting appropriate research strategies, methodologies, and data collection techniques..
CO3	differentiate between and apply quantitative and qualitative research methods appropriately based on the research question.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	-	-	-	-	-	-	-	-	-	1	-
CO2	1	2	1	-	-	-	-	-	-	-	-	-	1	-
CO3	1	2	2	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

1. C.R. Kothari and Gaurav Garg, “Research Methodology: Methods and Techniques”, New Age International (P) Ltd., Publishers, Fourth Multi Colour Edition, 2020.

❖ Reference Books:

1. Research Methodology And Statistical Tools - P. Narayana Reddy And G.V,R.K, Acharyulu, 1st Edition, Excel Books, New Delhi, 2008.

CE390: COMPETITIVE PROGRAMMING ESSENTIALS (PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	0	2	2
Marks	-	100	0	100	

Pre-requisite courses:

- Data Structures and Algorithms
- Familiarity with a programming language

Objectives of the Course:

- The primary objective of this course is to equip students with a comprehensive understanding of advanced algorithmic techniques essential for competitive programming and placement preparation. By mastering these techniques, students will not only enhance their problem-solving skills but also gain a competitive edge in coding competitions such as the ICPC.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to competitive programming and techniques	02
2.	Time Efficiency	04
3.	Mathematics for competitive programming	04
4.	Linear data structures and Libraries	05
5.	Non-linear data structures and Libraries	05
6.	Sorting, searching and algorithm design	06
7.	String processing algorithms	04
	Total hours (Theory) :	00
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	Introduction to competitive programming and techniques	02 Hours	07%
	Meaning Of Competitive Programming, Programming Techniques (Language Features Input/Output, Numbers, Code Shortening), Recursive Algorithms (Generating Subsets, Generating Permutations, Backtracking), Bit Manipulation (Bit Operations, Set Operations).		
2.	Time Efficiency	04 Hours	13%
	Common Time Complexity and Calculation Rules, Examples (Maximum Sub-Array Sum Problem, 2 Queens), Algorithm Analysis, Situational Awareness, Touch Typing Exercise to Improve Typing Speed.		
3.	Mathematics for competitive programming	04 Hours	13%
	Geometry (Points and Lines), Closest Pair Problem, Number Theory, Sum Formulae and Logarithms, Inclusion and Exclusion, Combinatorics		
4.	Linear data structures and Libraries	05 Hours	17%
	Static Array, Dynamic Array, Boolean Array, LinkedList, Stack, Queue, Double-Ended Queue, Exercise Problems.		
5.	Non-linear data structures and Libraries	05 Hours	17%
	Binary Search Tree (Map/Set), Heap (Priority Queue), Hash Table (Unordered Map), Graph (Adjacency List and Matrix, Edge List), Exercise Problems.		
6.	Sorting, searching and algorithm design	06 Hours	20%
	Complete Search (Iterative and Recursive), Linear and Binary Search, Sorting Algorithm and Library, Two Pointers, Nearest Smaller/Greater Element, Sliding Window, Ternary Search, Minimizing Sum, hamming distance, counting sub grids, reachability in graphs		
7.	String processing algorithms	04 Hours	13%
	Basic string processing, string processing problem categories, string matching, longest common subsequence, levenshtein distance		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	apply advanced algorithmic techniques to solve complex problems.
CO2	develop the ability to analyze and optimize the time complexity of algorithms.
CO3	gain mathematical skills crucial for algorithmic problem-solving
CO4	design efficient algorithms and develop strong problem-solving skills.
CO5	Compete in competitive programming contests and coding interviews.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	1	-	-	-	-	-	-	-	-	2	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	1	-	-	-	-	-	-	-	-	-	-	2	-
CO4	2	2	-	-	-	-	-	-	-	-	-	-	3	-
CO5	3	3	-	1	-	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. competitive programming by Steven & Felix Halim
2. Guide to Competitive Programming: Learning and Improving Algorithms Through Contests.

CE346: SUMMER INTERNSHIP-I

Credits and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours	90	-	-	90	3
Marks	150	-	-	150	

Objectives of the Course:

- To get familiar with modern tools and technologies use in company/industry/organization
- To get involved in design, development and testing practices followed in the company/industry/organization
- To enhance their soft-skills, presentation skills, interpersonal skills, documentation skills and office etiquettes required to sustain in company/industry/organization environment
- To participate in teamwork and preferably as part of a multi-disciplinary team
- To make them aware about company/industry/organization best practices, processes and regulations.
- To make them more productive, consistent and punctual.

Outline of the Course:

1.	Instructional Method and Pedagogy
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.

	• The student shall submit two documents to the Coordinator for the evaluation of the summer internship: • Summer Internship Report • Summer Internship Assessment Form • Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the Coordinator by the first day of the new term. • The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester. • If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.
2.	Format of Summer Internship Report
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers) • LIST OF FIGURES AND TABLES (with the corresponding page numbers)

- **DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION:**
Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- **INTRODUCTION:** In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it

	and waste tons of paper.	
	Total hours (Project) :	90
	Total hours :	90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ **Web material:**

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ **Software**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS
5. FLUTTER
6. NODE/REACT NATIVE

HS131.02 A: COMMUNICATION AND SOFT SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

Objectives of the Course:

- To hone and sharpen Communication Skills of students
- To prepare globally and multi-culturally competent communicators and professionally compatible cadre of future professionals
- To equip and empower students to qualify and successfully clear all the phases of selection procedure for on and off campus interviews

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization	03 Hours	10%

	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%
	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Text book:

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ **Reference book:**

1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ **Web material:**

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

HS142 A: French

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - French
- to learn basic grammar and sentence construction of French language
- to learn listening, speaking, reading and writing in French language
- to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to French Language	08
2.	French for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	26.66%
	- Facts and figures of French Language - Basic French Linguistics * Alphabets * Accents * Liaison * Nasalization - French Culture, Differ between French and English Grammar: ➤ Subject Pronoun ➤ Verbs: (être, avoir, habiter, regarder, manger ... “er” verb) ➤ Form of address ➤ Greetings ➤ Numbers (1 to 20) ➤ Nouns and plurals of nouns Ø The expression: C’est, Il y a Presentation: ➤ Self-Introduction ➤ Question and answering Dialogue		
2.	French for Basic Social Conversations	06 Hours	20%
	Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present tense ➤ Positive Forms ➤ Negative Forms ➤ Numbers (21 to 100, 100-1000) ➤ Days and Months Social Links: ➤ My family & relations ➤ Gathering information from someone Dialogue		
3.	Formal Conversations - I	06 Hours	20%
	Grammar: ➤ Common Adjectives ➤ Interrogative Forms ➤ Directions, Countries, Nationalities, Seasons ➤ Professions ➤ Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “verb) Work, Study and Travel Ø Job/ Profession		

	Ø Ticket Reservation (At Bus/At Railway/At Airport) □ Dialogue		
4.	Formal Conversations - II	06 Hours	20%
	• Grammar: ➤ Common Prepositions ➤ Common Conjunctions ➤ Future with verb “Aller” ➤ Colors, Shapes, Animals, Vegetables, Fruits • Food & Shopping ➤ Buy a vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) • Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Comparative Adjectives Ø Indiquer le lieu Ø The expression: “On” ➤ Irregular Verbs • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of French language and culture
CO2	use French for everyday social conversations
CO3	use basic French to talk in professional settings
CO4	talk in formal situations with basic French
CO5	use French for basic language functions

Evaluation:

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ **Text book:** -

❖ **Reference Reading:** The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with French 1 Platform:

The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-1/contentsection-overview?active-tab=description-tab>

2. Getting started with French 2 Platform:

The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-2/contentsection-overview?active-tab=description-tab>

3. Getting started with French 3

Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-3/contentsection-overview>

B. Tech. (Computer Engineering) Programme

SYLLABI (Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE364: CRYPTOGRAPHY & NETWORK SECURITY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Traditional Symmetric-Key Ciphers	04
3.	Introduction to Modern Symmetric-Key Ciphers	03
4.	Data Encryption Standard (DES)	03
5.	Advanced Encryption Standard (AES)	03
6.	Asymmetric-Key Cryptography	05
7.	Message Integrity and Message Authentication	04
8.	Cryptographic Hash Functions	04
9.	Digital Signature	04
10.	Entity Authentication	04
11.	Key Management	03
12.	Security at the Application Layer: PGP and S/MIME	02
13.	Security at the Transport Layer: SSL and TLS	02
14.	Security at the Network Layer: IPSec	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction	02 Hours	06%
	Security Goals, Attacks, Services and Mechanism and Techniques.		
2.	Traditional Symmetric-Key Ciphers	04 Hours	10%
	Introduction, Substitution Cipher, Transposition Cipher, Stream and Block Cipher		
3.	Introduction to Modern Symmetric-Key Ciphers	03 Hours	06%
	Modern Block Cipher, Modern Stream Cipher		
4.	Data Encryption Standard (DES)	03 Hours	10%
	Introduction, DES structure and Analysis, Multiple DES		
5.	Advanced Encryption Standard (AES)	03 Hours	06%
	Introduction, AES structure and Analysis		
6.	Asymmetric-Key Cryptography	05 Hours	10%
	Introduction, RSA Cryptosystem, RABIN Cryptosystem, ELGAMAL Cryptosystem		
7.	Message Integrity and Message Authentication	04 Hours	08%
	Message Integrity, Message Authentication		
8.	Cryptographic Hash Functions	04 Hours	08%
	Introduction, SHA-512,MD5		
9.	Digital Signature	04 Hours	06%
	Services,RSA Digital Signature Scheme,ELGamal Digital Signature Scheme		
10.	Entity Authentication	04 Hours	04%
	Passwords, Challenge –Response,Zero –Knowledge		
11.	Key Management	03 Hours	08%
	Symmetric-Key Distribution,KERBEROS		
12.	Security at the Application Layer: PGP and S/MIME	02 Hours	06%
	E-Mail, PGP,S/MIME		
13.	Security at the Transport Layer: SSL and TLS	02 Hours	06%
	SSL Architecture		
14.	Security at the Network Layer: IPSec	02 Hours	06%
	Introduction to IPSec, IPSec Tunneling		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Define various security goal and understand the security policies such as the CIA triad of Confidentiality, Integrity and Availability.
CO2	Classify various forms of security attacks, where they arise, and appropriate tools or mechanism to quantify them.
CO3	Illustrate a basic understanding of cryptography, how it has evolved, and evaluate symmetric key encryption techniques used today.
CO4	Distinguish modern symmetric encryption standard, key distribution scenario and analyse effectiveness in today's environment.
CO5	Evaluate Asymmetric key encryption techniques, key distribution scenario and calculate public and private components of asymmetric key encryption techniques.
CO6	Develop message integrity and message authentication of message digest.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

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Page **68** of **102**

1. Cryptography and Network Security, Behrouz A. Forouzan, McGraw-Hill Companies.

❖ **Reference book:**

1. Cryptography And Network Principles And Practice, William Stallings, Prentice Hall, Pearson Education Asia.
2. Cryptography & Network Security, Atul Kahate, The McGraw-Hill Companies.
3. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations by Gene Kim, Jez Humble, Patrick Debois, and John Willis

❖ **Web material:**

1. <http://people.csail.mit.edu/rivest/crypto-security.html>.
2. <http://www.cryptix.org/>
3. <http://www.cryptocd.org/>
4. <http://www.cryptopp.com/>
5. <http://www.freetechbooks.com/information-security-f52.html>

❖ **Software:**

1. Nmap
2. Wireshark

CE365: COMPILER CONSTRUCTION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics
- Operating System
- Theory of Computation

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Compiler	07
2.	Lexical Analysis	04
3.	Syntax Analysis	16
4.	Syntax directed translation (SDT)	04
5.	Semantic Analysis & Intermediate Code Generation	04
6.	Runtime Environment	05
7.	Code Optimization and Code Generation	05
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Introduction to Compiler	07 Hours	16%
	Overview of Language processor, Compiler and its types, Overview and Use of Linker and Loader, Phases of compiler, The science of building compilers [Front End and Back End of Compiler], Applications of language processors		
2.	Lexical Analysis	04 Hours	09%
	The role of lexical analyzer, Tokens-Patterns-Lexemes, Input Buffering Technique, specification of tokens – Strings and Languages, Regular Expressions, recognition of tokens, lexical		

	analyzer generator (LEX)		
3.	Syntax Analysis	16 Hours	35%
	The role of a parser, Context Free Grammar, Top-down parsing – Recursive-Descent Parsing, Predictive Parser, Non-recursive Predictive Parser, Bottom-up parsing – Shift reduce parsing, LR(k) Parser, LALR Parser, Operator Precedence Parsing, More powerful LR parsers, Using ambiguous grammars, Automatic Parser Generators - YACC		
4.	Syntax directed translation (SDT)	04 Hours	09%
	Syntax directed definitions (SDD), Construction of Syntax Tree, Evaluation order of SDD's – Bottom Up Evaluation of S-Attributed Definitions, L-Attributed Definitions, Syntax Directed Definitions and Translation Schemes, Applications of SDT		
5.	Semantic Analysis & Intermediate Code Generation	04 Hours	09%
	Intermediate Languages, Different Intermediate Forms – Abstract Syntax Tree, Polish Notation, Three Address Code and its implementation , Assignment Statements, Boolean Expressions, Back patching, Procedure Calls		
6.	Run Time Environment	05 Hours	11%
	Symbol Table - Data structures to implement symbol table, Symbol Attributes, Symbol-Table entries, Local Symbol Table management, Global Symbol Table Structure, Storage bindings and Symbolic Registers, Data descriptors - Static and Dynamic storage allocation – Storage allocation and access in block structured programming languages – Array allocation and access- Compilation of expressions – Handling operator priorities – Intermediate code forms for expressions – code generator., Register Usage, The run-time stack, Parameter passing disciplines, Code sharing and Position-Independent code		
7.	Code Optimization and Code Generation	05 Hours	11%
	The Principal Sources of Optimization, Optimization of Basic		

	Blocks, Loops in Flow Graphs, Introduction to Global Data-Flow Analysis, Iterative Solution of Data-Flow Equations, Code-Improving Transformations, Dealing with Aliases, Data-Flow Analysis of Structured Flow Graphs, Efficient Data-Flow Algorithms, A Tool for Data-Flow Analysis, Estimation of Types, Symbolic Debugging of Optimized Code, Issues in the Design of a Code Generator, The Target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Register Allocation and Assignment, The DAG Representation of Basic Blocks, Peephole Optimization, Generating Code from DAGs, Dynamic Programming Code-Generation Algorithm, Code-Generator Generators		
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Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand design and processing of different language processor, loaders and linkers
CO2	Design top-down and bottom-up parsers
CO3	Identify different memory management schemes of language processors
CO4	Develop semantic analysis scheme to generate intermediate code
CO5	Apply different code optimization techniques
CO6	Develop algorithms to generate code for a target machine

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	3	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	-	2	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, “Compilers Principles, Techniques and- Tools”, Pearson Education Asia.
2. M. Dhamdhare, “System Programming and Operating Systems”, Tata McGraw-Hill.
3. Steven S. Muchnick. Advanced Compiler Design and Implementation

❖ Reference book:

1. Allen I. Holub “Compiler Design in C”, Prentice Hall of India.
2. C. N. Fischer and R. J. LeBlanc, “Crafting a compiler with C”, Benjamin Cummings.
3. J.P. Bennet, “Introduction to Compiler Techniques”, Second Edition, Tata McGraw-Hill
4. HenkAlblas and Albert Nymeyer, “Practice and Principles of Compiler Building with C”, PHI.
5. Kenneth C. Loudon, “Compiler Construction: Principles and Practice”, Thompson Learning.
6. Compiler Construction by Kenneth. C. Loudon, Vikas Pub

❖ Web material:

1. <http://compilers.iecc.com/crenshaw>
2. <http://www.compilerconnection.com>
3. <http://dinosaur.compilertools.net>
4. <http://pltplp.net/lex-yacc>

❖ Software:

1. LEX
2. YACC

CE366: CLOUD COMPUTING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Operating System, Networking, Distributed System

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08
2.	Fundamental of Virtualization	06
3.	Fundamental Concepts and Models	06
4.	Cloud-Enabling Technology	05
5.	Fundamental Cloud Architectures	07
6.	Advanced Cloud Architectures	08
7.	Implementation of Cloud	05
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08 Hours	18%
	Introduction, Eras of Computing, Scalable Parallel Computer Architectures, Towards Low Cost Parallel Computing and Motivations , A Cluster Computer and its Architecture, Clusters Classification, Commodity Components for Clusters, Grid Architecture and Service Modelling, Grid Projects and Grid Systems Built, Grid Resource Management and		

	Brokering, Software and Middleware for Grid Computing, Grid Application Trends		
2.	Fundamental of Virtualization	06 Hours	12%
	Type of Virtualization, Virtualization Technologies, Virtualizes your Environment, Managing Virtualization Environment, Storage Virtualization, Dockers		
3.	Fundamental Concepts and Models	06 Hours	15%
	Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models		
4.	Cloud-Enabling Technology	05 Hours	14%
	Broadband Networks and Internet Architecture, Data centre Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology		
5.	Fundamental Cloud Architectures	07 Hours	16%
	Workload Distribution Architecture, Resource Pooling Architecture ,Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture		
6.	Advanced Cloud Architectures	08 Hours	17%
	Hypervisor Clustering Architecture ,Load Balanced Virtual Server Instances Architecture, Non-Disruptive Service Relocation Architecture, Zero Downtime Architecture ,Cloud Balancing Architecture ,Resource Reservation Architecture, Dynamic Failure Detection and Recovery Architecture		
7.	Implementation of Cloud	05 Hours	08%
	Study of Cloud Computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools, Deployment of Web Services from Inside and Outside a Cloud Architecture. MapReduce and its extensions to Cloud Computing, HDFS, and GFS		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Assess and examine advantages and disadvantages of virtualization technology.
CO2	Compose services in a distributed computing environment to achieve tasks relevant to a knowledge-based business or public service
CO3	Evaluate a set of business requirements to determine suitability for a cloud computing delivery model.
CO4	Explore the various cloud computing architectures and paradigms.
CO5	Deployment of cloud and identify security implications in cloud computing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	1	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO3	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO4	3	-	3	2	-	-	-	-	-	-	-	-	1	-
CO5	3	-	3	2	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:**❖ Text book:**

1. Thomas Erl, Zaigham Mahmood, and Ricardo Puttini, “Cloud Computing Concepts, Technology & Architecture”, Prentice Hall
2. Kai Hwang, Geoffrey C.,” Distributed and Cloud Computing”, Morgan Kaufmann is an imprint of Elsevier
3. Navin Sabharwal, Ravi Shankar “Apache CloudStack Cloud Computing” PACKT Publishing

❖ **Reference book:**

1. Ravi Shankar, Navin Sabharwa “Cloud Computing First Steps: Cloud Computing for Beginners” Create Space Independent Publishing Platform
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski “Cloud Computing: Principles and Paradigms” Wiley
3. Judith Hurwitz, Robin Bloor “Cloud Computing For Dummies”, for Dummies

❖ **Web material:**

1. <http://www.console.cloud.google.com>
2. <http://www.qwicklabs.com>
3. <http://codelabs.developers.google.com>
4. <http://www.docker.com>

❖ **Software/Platform:**

1. Amazon AWS
2. Microsoft
3. NetBeans
4. Eclipse
5. .NET
6. GCP

OCCE3001: DEEP LEARNING (PE-III)

Description:

This course OCCE3001– Deep Learning is offered from SWAYAM as noc24_cs59 – Deep Learning

URL : https://onlinecourses.nptel.ac.in/noc24_cs59/preview

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

* Practical component is offered by CHARUSAT

About this course:

Deep Learning has received a lot of attention over the past few years and has been employed successfully by companies like Google, Microsoft, IBM, Facebook, Twitter etc. to solve a wide range of problems in Computer Vision and Natural Language Processing. In this course, we will learn about the building blocks used in these Deep Learning based solutions. Specifically, we will learn about feedforward neural networks, convolutional neural networks, recurrent neural networks and attention mechanisms. We will also look at various optimization algorithms such as Gradient Descent, Nesterov Accelerated Gradient Descent, Adam, AdaGrad and RMSProp which are used for training such deep neural networks. At the end of this course, students would know deep architectures used for solving various Vision and NLP tasks

Course Layout:

Week 1: (Partial) History of Deep Learning, Deep Learning Success Stories, McCulloch Pitts Neuron, Thresholding Logic, Perceptrons, Perceptron Learning Algorithm

Week 2: Multilayer Perceptrons (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, Feedforward Neural Networks, Representation Power of Feedforward Neural Networks

Week 3: FeedForward Neural Networks, Backpropagation

Week 4: Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam, Eigenvalues and eigenvectors, Eigenvalue Decomposition, Basis.

Week 5: Principal Component Analysis and its Interpretations, Singular Value Decomposition

Week 6: Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders.

Week 7: Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout.

Week 8: Greedy Layerwise Pre-training, Better activation functions, Better weight initialization methods, Batch Normalization.

Week 9: Learning Vectorial Representations Of Words

Week 10: Convolutional Neural Networks, LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks.

Week 11: Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs.

Week 12: Encoder-Decoder Models, Attention Mechanism, Attention over images

Books and References:

- Deep Learning, An MIT Press book, Ian Goodfellow and Yoshua Bengio and Aaron Courville <http://www.deeplearningbook.org>

CRITERIA TO GET A CERTIFICATE:

Average assignment score = 25% of the average of the best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

Once again, thanks for your interest in our online courses and certification.

OCCE3002: BIG DATA COMPUTING (PE-III)

Description:

This course OCCE3002– Big Data Computing is offered from SWAYAM as noc23_cs83 – Big Data Computing

URL : https://onlinecourses.nptel.ac.in/noc23_cs83/preview

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

* Practical component is offered by CHARUSAT

About this course:

In today's fast-paced digital world , the incredible amount of data being generated every minute has grown tremendously from sensors used to gather climate information, posts to social media sites, digital pictures and videos, purchase transaction records, and GPS signals from cell phone to name a few. This amount of large data with different velocities and varieties is termed as big data and its analytics enables professionals to convert extensive data through statistical and quantitative analysis into powerful insights that can drive efficient decisions. This course provides an in-depth understanding of terminologies and the core concepts behind big data problems, applications, systems and the techniques, that underlie today's big data computing technologies. It provides an introduction to some of the most common frameworks such as Apache Spark, Hadoop, MapReduce, Large scale data storage technologies such as in-memory key/value storage systems, NoSQL distributed databases, Apache Cassandra, HBase and Big Data Streaming Platforms such as Apache Spark Streaming, Apache Kafka Streams that has made big data analysis easier and more accessible. And while discussing the concepts and techniques, we will also look at various applications of Big Data Analytics using Machine Learning, Deep Learning, Graph Processing and many others. The course is suitable for all UG/PG students and practicing

engineers/ scientists from the diverse fields and interested in learning about the novel cutting edge techniques and applications of Big Data Computing.

Course Layout:

Week 1: Introduction to Big Data

Week 2: Introduction to Enabling Technologies for Big Data

Week 3: Introduction to Big Data Platforms

Week 4: Introduction to Big Data Storage Platforms for Large-Scale Data Storage

Week 5: Introduction to Big Data Streaming Platforms for Fast Data

Week 6: Introduction to Big Data Applications (Machine Learning)

Week 7: Introduction of Big data Machine learning with Spark

Week 8: Introduction to Big Data Applications (Graph Processing)

Books and References:

- Dirk Deroos et al., Hadoop for Dummies, Dreamtech Press, 2014.
- Chuck Lam, Hadoop in Action, December, 2010.
- Leskovec, Rajaraman, Ullman, Mining of Massive Datasets, Cambridge University Press.
- I.H. Witten and E. Frank, Data Mining: Practical Machine learning tools and techniques.
- Erik Brynjolfsson et al., The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies, W. W. Norton & Company, 2014.

CRITERIA TO GET A CERTIFICATE:

Average assignment score = 25% of average of best 6 assignments out of the total 8 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

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at nptel.ac.in/noc.

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Once again, thanks for your interest in our online courses and certification

OCCE3003: INTRODUCTION TO INTERNET OF THINGS (PE-III)

Description:

This course OCCE3003–Introduction to Internet of Things is offered from SWAYAM as noc23_cs83 – Introduction to Internet of Things

URL : https://onlinecourses.nptel.ac.in/noc23_cs83/preview

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

* Practical component is offered by CHARUSAT

About this course:

The Internet of Things (IoT) is presently a hot technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT. IoT cuts across different application domain verticals ranging from civilian to defence sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support IoT. Today it is possible to envision pervasive connectivity, storage, and computation, which, in turn, gives rise to building different IoT solutions. IoT-based applications such as innovative shopping systems, infrastructure management in both urban and rural areas, remote health monitoring and emergency notification systems, and transportation systems, are gradually relying on IoT-based systems. Therefore, it is very important to learn the fundamentals of this emerging technology..

Course Layout:

Week 1: Introduction to IoT: Part I, Part II, Sensing, Actuation, Basics of Networking: Part I

Week 2: Basics of Networking: Part II, Part III, Part IV, Communication Protocols: Part I, Part II

Week 3: Communication Protocols: Part III, Part IV, Part V, Sensor Networks: Part I, Part II

Week 4: Sensor Networks: Part III, Part IV, Part V, Part VI, Machine-to-Machine Communications

Week 5: Interoperability in IoT, Introduction to Arduino Programming: Part I, Part II, Integration of Sensors and Actuators with Arduino: Part I, Part II

Week 6: Introduction to Python programming, Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi

Week 7: Implementation of IoT with Raspberry Pi (contd), Introduction to SDN, SDN for IoT

Week 8: SDN for IoT (contd), Data Handling and Analytics, Cloud Computing

Week 9: Cloud Computing(contd), Sensor-Cloud

Week 10: Fog Computing, Smart Cities and Smart Homes

Week 11: Connected Vehicles, Smart Grid, Industrial IoT

Week 12: Industrial IoT (contd), Case Study: Agriculture, Healthcare, Activity Monitoring

Books and References:

- S. Misra, A. Mukherjee, and A. Roy, 2020. *Introduction to IoT*. Cambridge University Press.
- S. Misra, C. Roy, and A. Mukherjee, 2020. *Introduction to Industrial Internet of Things and Industry 4.0*. CRC Press.
- Research Papers

CRITERIA TO GET A CERTIFICATE:

Average assignment score = 25% of the average of the best 6 assignments out of the total 8 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF THE AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

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Only the e-certificate will be made available. Hard copies will not be dispatched.

OCCE3004: BLOCKCHAIN AND ITS APPLICATIONS (PE-III)

Description:

This course OCCE3004– Blockchain and its Applications is offered from SWAYAM as noc24_cs15 – Blockchain and its Applications

URL : https://onlinecourses.nptel.ac.in/noc24_cs15/preview

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

* Practical component is offered by CHARUSAT

About this course:

In the last few years, Blockchain technology has generated massive interest among governments, enterprises, and academics, because of its capability of providing a transparent, secured, tamper-proof solution for interconnecting different stakeholders in a trustless setup. In January 2021, the Ministry of Electronics and Information Technology (MeiTY), Government of India, published the first draft of the "National Strategy on Blockchain" that highlights 17 potential applications that are of national interest. Against this backdrop, this subject will cover the basic design principles of Blockchain technology and its applications in different sectors. Additionally, the course also provides tutorials on setting up blockchain applications using one of the well-adopted permissionless blockchain platforms - Ethereum, and one permission blockchain platform - Hyperledger.

Course Layout:

Week 1: Introduction to Blockchain Technology and its Importance

Week 2: Basic Crypto Primitives I – Cryptographic Hash

Week 3: Basic Crypto Primitives II – Digital Signature

Week 4: Evolution of the Blockchain Technology

Week 5: Elements of a Blockchain

Week 6: Blockchain Consensus I – Permissionless Models

Week 7: Blockchain Consensus II – Permissioned Models

Week 8: Smart Contract Hands On I – Ethereum Smart Contracts (Permissionless Model)

Week 9: Smart Contract Hand On II – Hyperledger Fabric (Permissioned Model)

Week 10: Decentralized Identity Management

Week 11: Blockchain Interoperability

Week 12: Blockchain Applications

Books and References:

- Mastering Blockchain: A deep dive into distributed ledgers, consensus protocols, smart contracts, DApps, cryptocurrencies, Ethereum, and more, 3rd Edition, Imran Bashir, Packt Publishing, 2020, ISBN: 9781839213199, book website: <https://www.packtpub.com/product/mastering-blockchain-third-edition/9781839213199>
- Hyperledger Tutorials - <https://www.hyperledger.org/use/tutorials>
- Ethereum Development Resources - <https://ethereum.org/en/developers>
- Online materials and case studies

CRITERIA TO GET A CERTIFICATE:

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Kharagpur. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

Once again, thanks for your interest in our online courses and certification.

OCCE3005: ADVANCED COMPUTER NETWORKS (PE-III)

Description:

This course OCCE3005– Advanced Computer Networks is offered from SWAYAM as noc24_cs11 – Advanced Computer Networks

URL : https://onlinecourses.nptel.ac.in/noc24_cs11/preview

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

* Practical component is offered by CHARUSAT

About this course:

This is an advanced course on Computer Networks covering four main contemporary topics namely how high-performance networks are designed covering algorithms and implementations, data centre networking, information-centric networking and network softwarization: software-defined networking, network virtualization and programmable networks. This course assumes a basic knowledge of computer networks and working proficiency with Unix/Linux Operating Systems. It has practical components with programming assignments given hands-on to the students.

Course Layout:

Week 1: High Performance Switching and Routing: Introduction, performance considerations, IP address lookup

Week 2: Algorithms for IP address lookup and optimization, hardware implementation of address lookup

Week 3: Packet Classification: Need for packet classification and methods for packet classification.

Week 4: Differentiated Service, Quality of Service, Traffic Polishing, Traffic Shaping

Week 5: Network Softwarization - Introduction

Week 6: Software Defined Networking (SDN) - Deep Dive (Northbound and Southbound interface), Working with Mininet + Lab Exercises with Mininet

Week 7: Network Function Virtualization (NFV) - Architecture and Concepts

Week 8: Programmable Networks - Introduction to P4, SmartNICS and P4 switches. + Lab Exercise with Mininet and BMV2 switches.

Week 9: Data Center Networking (DCN) - Introduction

Week 10: DCN - Deep Dive (Network topologies, Container Network Interfaces)

Week 11: Content Distribution on the Internet, Architectures for Information-Centric Networking

Week 12: Content Naming, Routing and Caching, Security in Named Data Networking

Books and References:

- High-Performance Switches and Routers, H. Jonathan Chao, Bin Liu, 2007, John Wiley & Sons, Inc. ISBN-10: 0-470-05367-4
- Information-Centric Networks: A New Paradigm for the Internet (Focus Series in Networks and Telecommunications), Gabriel M. de Brito, Pedro B. Velloso, Igor M. Moraes, Wiley-ISTE; 1st edition, 2013, ISBN: 9781848214491
- Information-Centric Networking (ICN): Content-Centric Networking (CCNx) and Named Data Networking (NDN) Terminology, B. Wissingh, C. Wood, A. Afanasyev, L. Zhang, D. Oran and C. Tschudin, RFC 8793, June 2020
- Software-Defined Networks: A Systems Approach, Peterson, Cascone, O'Connor, Vachuska, and Davie, Online Free Reference Book available at <https://sdn.systemsapproach.org/index.html>
- Cloud Networking: Understanding Cloud-based Data Centre Networks, Gary Lee (Author), Morgan Kaufmann (Publisher), 2014, ISBN-139780128007280 Hyperledger Tutorials - <https://www.hyperledger.org/use/tutorials>

CRITERIA TO GET A CERTIFICATE:

Average assignment score = 25% of the average of the best 8 assignments out of the total 12 assignments given in the course.

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF THE AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND THE EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

A certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

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CE367: PROJECT - IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	4
Marks	-	200	-	200	

Pre-requisite courses:

- Programming Knowledge

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on a weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
 - Project Synopsis
 - Software Requirement Specification
 - SPMP
 - Final Project Report
 - Project Setup file with Source code
 - Project Presentation (PPT)

- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 60

Total hours : 60

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Prepare team formation strategies and stages leading to the development of high performing, self-managing teams.
CO2	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply engineering and management principles to achieve project goal.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Recognition of the need for and an ability to engage in continuing professional development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	-	-	-	-	-	-	3	-	-	1	-	2
CO2	-	3	1	2	-	-	-	-	-	-	-	-	2	-
CO3	3	-	2	-	1	-	-	-	-	-	-	-	2	-
CO4	3	-	-	2	-	-	1	-	-	-	-	-	3	-
CO5	-	-	-	-	2	-	-	-	-	1	3	-	-	2
CO6	-	-	-	-	-	1	-	2	-	-	-	3	-	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Books, Magazines & Journals of related topics

❖ Web material:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ Software:

1. ASP.NET
 2. PYTHON/MATLAB
 3. PHP
 4. ANDROID/IOS
-

CE396: RESEARCH WRITING AND ETHICS (PE-IV)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Pre-requisite courses:

Critical Thinking and Analytical Skills, Fundamentals of Academic Writing.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Research Problem Formulation	06
2.	Research Paper Writing	08
3.	Databases and Research Metrics	06
4.	Research Ethics	10
	Total hours (Theory) :	-
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	Research Problem Formulation	06 Hours	25%
	Defining and Formulating the Research Problem: Selecting the Problem, Necessity of Defining the Problem, Technique Involved in Defining a Problem.		
2.	Research Paper Writing	08 Hours	30%
	Types of research papers, Structure of research papers, Research paper formats, Abstract writing, Methodology, Results and discussions., Steps to Organize Manuscript, Length of Research Article		
3.	Databases and Research Metrics	06 Hours	15%

	Definition of Databases, Indexing databases, Citation databases: Web of Science, Scopus, etc. Definition of Research Metrics - Impact Factor of journal as per Journal Citation Report, SNIP, SJR, IPP, Cite Score, Metrics: h-index, g index, i10 index.		
4.	Research Ethics	10 Hours	30%
	Definition of Plagiarism; Consequences of Plagiarism; Unintentional Plagiarism; Forms of Plagiarism. Peer Review Process, Publication Ethics		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Conduct a thorough literature review, synthesize existing research, and identify gaps in the current knowledge.
CO2	Understand and apply ethical principles in conducting research, including obtaining informed consent, ensuring confidentiality, and avoiding plagiarism..
CO3	write clear and well-structured research papers and reports, and effectively present their research findings to various audiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	-	-	-	-	-	-	-	-	-	1	-
CO2	1	2	2	-	-	-	-	-	-	-	-	-	1	-
CO3	1	3	2	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text Books:

- American Psychological Association (2010) Publication Manual of the American Psychological Association (6th ed.). Washington, DC: APA.

❖ Reference Books:

1. Day, R. and Castel, B. (2012). How to Write and Publish a Scientific Paper (7thed.). Cambridge: Cambridge University Press.
2. Research Ethics: A Psychological Approach By Barbara H. Stanley; Joan E. Sieber; Gary B.Melton.

CE397: MASTERING COMPETITIVE PROGRAMMING (PE-IV)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	0	2	2
Marks	-	100	0	100	

Pre-requisite courses:

- Data Structures and Algorithms

Objectives of the Course:

- The primary objective of this course is to equip students with a comprehensive understanding of advanced algorithmic techniques essential for competitive programming and placement preparation. By mastering these techniques, students will not only enhance their problem-solving skills but also gain a competitive edge in coding competitions such as the ICPC.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Range Queries	04
2.	Algorithms on tree data structure	05
3.	Algorithms on graph data structure – 1	04
4.	Algorithms on graph data structure – 2	04
5.	Greedy programming	04
6.	Dynamic programming	06
7.	Additional topics	03
	Total hours (Theory):	00
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	Range Queries	04 Hours	12%
	Queries on static arrays (sum and minimum queries), tree structures (binary indexed trees, segment trees), additional techniques		
2.	Algorithms on tree data structure	05 Hours	15%
	Basic techniques (traversal, diameter, all longest paths), tree queries (find ancestors, subtrees, lowest common ancestors)		
3.	Algorithms on graph data structure – 1	04 Hours	12%
	Basic of graph (graph terminologies, representation), graph traversal (DFS, BFS, applications), revision of shortest paths, successor graphs, MST,		
4.	Algorithms on graph data structure – 2	04 Hours	20%
	Traversal problems (finding connected components, flood fill, bipartite graph check, articulation points and bridges, strongly connected components), single source shortest paths, all pair shortest paths, special graphs (DAG, eulerian, bipartite)		
5.	Greedy programming techniques	04 Hours	12%
	Coin change revisited, load balancing, interval covering, sort the input first		
6.	Dynamic programming	06 Hours	20%
	Finding an optimal solution, longest increasing subsequence, paths in grid, permutations to subsets, counting tilings, classical examples		
7.	Additional topics	03 Hours	9%
	Square root techniques, segment trees, dynamic programming optimization, miscellaneous		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Solve and apply concept of tree and graph data structure for solving real time problems
CO2	Apply greedy algorithms to solve optimization problem
CO3	Apply dynamic programming techniques for solving complex problems.
CO4	design efficient algorithms and develop strong problem-solving skills.
CO5	Compete in competitive programming contests and coding interviews.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	2	-	-	-	-	-	-	-	-	-	-	3	-
CO5	3	3	-	1	-	-	-	-	-	-	-	-	2	-

Recommended Study Material:**❖ Text book:**

3. competitive programming by Steven & Felix Halim
4. Guide to Competitive Programming: Learning and Improving Algorithms Through Contests.

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

Objectives of the Course:

- Become familiar with basic concept of personality and personality development
- Understand personality development theories and strategies
- Evaluate one's personality and inculcate traits of an assertive personality
- Develop an assertive personality
- Develop life skills and required management traits
- Enhance contributory personality for academic and career success

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		

3.	Developing Contributory Personality	06 Hours	20%
	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Text book:

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ Reference book:

1. Contributor Personality Program Workbook (Volume 1,2),

2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

1. <https://www.coursera.org/learn/wharton-success>
2. <https://www.coursera.org/learn/personality-types-at-work>
3. <https://www.coursera.org/learn/self-awareness>